

ASTRODYNAMICS

The TANGO-Sat mission's trajectory to Low Earth Orbit was designed to fully utilize the performance, reliability, and direct-injection capabilities of SpaceX's Falcon 9 launch vehicle. The overall trajectory design reflects two major mission phases: ascent and orbital insertion into the target sun-synchronous orbit, followed by the end-of-life controlled disposal sequence. Both phases were modeled in Systems Tool Kit (STK) using finite-thrust propagators to accurately represent the spacecraft's electric propulsion behavior, the continuous low-thrust nature of the burns, and the non-impulsive dynamics that characterize small-satellite propulsion systems (Systems Tool Kit (STK) Astrogator Module User Guide, 2024).

Falcon 9 provides substantial lift capability, delivering up to 22,800 kg to Low Earth Orbit and reliability supporting direct insertion into high-inclination sun-synchronous trajectories when launched from Vandenberg Space Force Base (Space Exploration Technologies, 2025). Leveraging Falcon 9's performance eliminates the need for any onboard plane-change maneuver, a process typically very ΔV -intensive for small satellites, and significantly simplifies the propulsion, powder, and operational design of the TANGO-Sat system. By allowing the launch vehicle to provide nearly all the required energy for orbital geometry, mission resources can instead be allocated to payload operations and safe, compliant disposal.

Phase 1: Launch and LEO insertion

The TANGO-Sat mission launches aboard SpaceX Falcon 9 from Vandenberg Space Force Base (SLC-4E), a site uniquely suited for sun-synchronous missions due to its open southbound polar launch corridor. This range provides direct access to high-inclination orbits without requiring an overflight of populated areas, supporting both safety and regulatory requirements. The primary launch window opens on 14 September 2027 at 4 P.M. local time and remains available for 16 minutes, a duration that reflects the precise orbital alignment needed for a clean injection into the mission's sun-synchronous geometry. During this window, the Earth's rotation brings the launch site into alignment with the spacecraft's descending node, allowing the Falcon 9 to ascend directly into the target plane without additional costly plane-change maneuvers. Launching at the descending node also ensures that, in the unlikely event of an off-nominal breakup or early mission failure, any debris would follow a trajectory that safely impacts the Pacific Ocean rather than populated areas.

This timing requirement is not flexible; it stems from the repeating relationship between the Earth's rotation and the position of the orbital plane. Even slight deviations in launch time shift the required launch azimuth, potentially exceeding the allowable performance margins for the vehicle or conflicting with safety constraints established for polar launches. As a result, the 16-minute window represents the narrow period in which all geometric, safety, and performance considerations align, enabling a reliable and efficient insertion into the mission's required orbit.

A scrub opportunity becomes available 24 hours later, aligned with the next descending-node crossing. Launching at this node also ensures that any debris from an off-nominal event, such as a breakup or early mission failure, would fall safely into the Pacific Ocean, consistent with range safety guidelines. The mid-September timeframe further benefits from historically stable weather at Vandenberg, low upper-level wind shear, and minimal lightning constraints, all of which contribute to high launch probability and reduced schedule risk.

Following liftoff, Falcon 9's first and second stages operate for a combined duration of 540 seconds, accelerating the vehicle to a burnout altitude near 500 km with a launch azimuth of -169.98° and a downrange distance of 17,575.3 km (Space Launch Report: SpaceX Falcon 9 Data Sheet, 2017). Booster recovery occurs offshore after first-stage separation, while the second stage completes the ascent with a final injection burn. Once the payload fairing separates around 80 km altitude, the second stage conducts the terminal guidance, navigation, and control sequence required to insert the spacecraft directly into a 500 km, 97° sun-synchronous orbit, placing TANGO-Sat immediately in its operational configuration with no orbit-raising required.

Phase 2: deorbit and disposal

TANGO-Sat follows a controlled end-of-life deorbit strategy to ensure compliance with NASA and international debris-mitigation requirements. Starting from the operational 500 km, 97° sun-synchronous orbit, the spacecraft performs a two-burn Hohmann transfer to lower its perigee to 300 km. The required ΔV of 112.4 m/s is split between a 57.99 m/s retrograde burn at apoapsis and a 54.5 m/s burn at periapsis, with detailed calculations outlined in Appendix B. Burns were modeled using finite-thrust electric propulsion in STK, producing burn durations of 1 hour and 26 minutes (ΔV_1 at apoapsis) and 1 hour and 21 minutes (ΔV_2 at periapsis). Including a 20% propulsion margin, the total

propellant allocated is 2.71 kg, calculated from the rocket equation.

At 300 km, atmospheric drag governs the orbital decay process. While the predicted natural lifetime of approximately 252 days remains within regulatory limits, the decay timeline is highly sensitive to uncertainties in solar activity, density modeling, and the spacecraft's attitude and ballistic coefficient. These uncertainties preclude the level of precision required for disposal targeting at Point Nemo.

The final targeting sequence consists of two maneuvers: a controlled deorbit burn and a subsequent phasing correction for Point Nemo targeting. First, a small finite retrograde burn lowers the perigee to 80 km to ensure destructive atmospheric breakup. Using the vis-viva equation, detailed in Appendix B, this maneuver requires a ΔV of 65.07 m/s, corresponding to a propellant mass of 1.34 kg and a burn time of 1 hour and 38 minutes. Following this burn, 1.95 kg of propellant remains available for phasing and margin.

To align the ground track so that re-entry occurs over Point Nemo (-48.52° latitude, -123.23° longitude), a 90° phasing adjustment distributed over roughly 1,000 orbits (about 65 days) requires an additional ΔV of 10.24 m/s, with an associated propellant mass of 0.214 kg, and a burn duration of approximately 15 minutes. A 20% performance margin is included to accommodate uncertainties in attitude control, atmospheric density variability, and navigation during late-stage decay (Deorbit Systems, 2021). Although relatively small, these maneuvers are essential, as natural drag decay alone cannot provide the precision needed to target Point Nemo. Executing this sequence ensures compliance with the End-of-Mission Disposal Requirement by guaranteeing destructive re-entry over an unpopulated region.

Inclination and Altitude Reasoning

The mission targets a 97° sun-synchronous orbit (SSO) to provide consistent lighting, thermal, and operational conditions throughout its six-month duration. A sun-synchronous inclination ensures that each ground track occurs at nearly the same local solar time, yielding repeatable illumination for imaging, predictable eclipse periods, and stable thermal and power environments for spacecraft planning. This orbital geometry also improves communications performance by offering highly regular, predictable pass times which is particularly advantageous for high-latitude or polar ground stations. The main ground station is located in Svalbard, Norway near the city of Longyearbyen. At a 97° inclination at a 500 km altitude enables 14 passes over this ground

station per day with a duration of about 10 minutes per pass.

The corresponding tradeoffs, such as increased ΔV requirements due to the retrograde component of a 97° inclination and fewer long-duration daytime passes at lower latitudes, were accepted based on the operational benefits and the launch vehicle's capability to inject directly into this orbit.

The mission altitude was chosen as 500 km to eliminate the need for station-keeping maneuvers, which would be difficult or infeasible to execute while the tethered system is spinning. At 400 km, atmospheric drag would induce significant orbital decay over the mission lifetime, requiring multiple corrective burns including multi-impulse sequences to maintain inclination thus complicating operations and increasing risk. By selecting a 500 km altitude, the spacecraft avoids meaningful decay over the six-month mission without requiring orbit maintenance. The Falcon 9 can deliver the spacecraft directly into the 97° and 500 km orbit, and the additional insertion cost has been incorporated into the mission budget, removing the need for an onboard plane-change maneuver and simplifying propulsion system design.

REFERENCES

- BASF SE. (2016). *Basotect®: The versatile melamine resin foam*. (BASF) Retrieved December 2, 2025, from https://www.basf.com/dam/jcr%3A142883a0-d419-3e59-8b1f-2698c39c5ccd/basf/www/us/documents/en/Basotect_Br_e_ab02_2016.pdf
- Buitrago-Leiva, J. N., El Khayati, R. M., Camps, A., & Ruiz-de-Azua, J. A. (2024). Statistical Analysis of LEO and GEO Satellite Anomalies and Space Radiation. *Aerospace*.
- CubeSpace Satellite Systems *CubeSense: Fine Sun Sensor*. (n.d.). Retrieved from CubeSpace: <https://www.cubespace.co.za/products/fine-sun-sensor/?carousel=true>
- Custom Thermoelectric. (2025). *Thermoelectric/Peltier Mini Module*. Retrieved 2025, from <https://customthermoelectric.com/04801-9a30-18rb.html>
- Deorbit Systems. (2021, October). (NASA) Retrieved 2025, from <https://www.nasa.gov/smallsat-institute/sst-soa/deorbit-systems/>
- Dwyer Omega. (2025). *Polyimide Flexible Heater*. (Dwyer Omega) Retrieved 2025, from <https://www.dwyeromega.com/en-us/polyimide-flexible-heater-sample-kit-15-different-heaters/p/KHA-KIT-Series>
- Experior Laboratories. (n.d.). (Experior Laboratories, LLC) Retrieved from <https://experiorlabs.com/>
- Finckenor, M. M. (n.d.). *Multilayer Insulation Material Guidelines*. Retrieved 2025, from <https://ntrs.nasa.gov/api/citations/19990047691/downloads/19990047691.pdf>
- Fitz-coy, N. G. (2025). *Notes on the Dynamics and Control of Space Vehicles*. unpublished.
- LISEGA Group. (2020, May). *Product Catalogue: Viscoelastic Dampers*. (LISEGA Group) Retrieved December 2, 2025, from https://www.lisega.de/wp-content/uploads/2020/05/Product_catalogue_Viskodaempfer_210x297_en_2020.pdf
- Mouser Electronics. (2025). *Honeywell*. Retrieved 2025, from <https://www.mouser.com/ProductDetail/Honeywell/HEL-705-U-0-12-00?qs=%2Ffq2y7sSKcIQVzFb76FF2A%3D%3D&mgh=1&utm>
- NASA. (2025, February 5). *State-of-the-Art of Small Spacecraft Technology*. Retrieved 2025, from <https://www.nasa.gov/smallsat-institute/sst-soa/thermal-control/#7.3>
- NASA Goddard Space Flight Center. (2021, April 28). *General Environmental Verification Standard (GEVS) for GSFC Flight Programs and Projects*. (Nasa) Retrieved december 2, 2025
- Peabody, H. (2022). Thermal Design for Spaceflight.
- Prasad, B. B., Duvigneau, F., Juhre, D., & Woschke, E. (2022). Damping performance of particle dampers with different granular materials and their mixtures. *Applied Acoustics*, 200, 109059.
- Space Exploration Technologies. (2025). *Falcon User's Guide*.
- Space Launch Report: *SpaceX Falcon 9 Data Sheet*,. (2017). (NASA) Retrieved 2025, from <https://sma.nasa.gov/LaunchVehicle/assets/spacex-falcon-9-data-sheet.pdf>
- Space X. (March 2025). *Falcon User's Guide, Version 8*.
- SpaceManic *Celeste gnss_rtx*. (n.d.). Retrieved from Spacemaniac: <https://spacemanic.com/products/celeste/>
- Systems Tool Kit (STK) *Astrogator Module User Guide*. (2024). (AGI/Ansys) Retrieved 2025, from <https://www.agi.com/products/stk>
- University of Florida. (2025). *Space Weather Advanced Monitoring Platform Satellite (GATOR-SAT) Requirements Document*. Gainesville, FL.
- University of Florida. (n.d.). *Smart Infrastructure Management Laboratory*. Retrieved 2025, from <https://simlab.essie.ufl.edu/facilities/>
- VectorNav *VN-100*. (n.d.). Retrieved from VectorNav: <https://www.vectornav.com/products/detail/vn-100>
- ZARM Technik. (n.d.). Retrieved from ZARM Technik: <https://www.zarm-technik.de/>

APPENDIX B: ASTRODYNAMICS

Below is the Tsiolkovsky Rocket Equation is used to determine the fuel mass necessary given a ΔV value

$$\Delta V = I_{sp} g_0 \ln \left(\frac{m_0}{m_f} \right)$$

where m_0 is the initial (wet) mass, m_f is the final (dry) mass, I_{sp} is the specific impulse, and g_0 is the earth's gravitational acceleration.

The ΔV values necessary to perform the Hohmann Transfer de-orbit burns are found using the following equations

$$\Delta v_1 = \sqrt{\frac{\mu}{r_1}} \left(\sqrt{\frac{2r_2}{r_1 + r_2}} - 1 \right)$$

$$\Delta v_2 = \sqrt{\frac{\mu}{r_2}} \left(1 - \sqrt{\frac{2r_1}{r_1 + r_2}} \right)$$

$$\Delta v_{Hohmann} = \Delta v_1 + \Delta v_2$$

where Δv_1 is the ΔV required to leave a circular orbit and enter a transfer ellipse and Δv_2 is the ΔV required to circularize the new orbit.

The ΔV values necessary to perform the targeting deorbit burns are found using the following equations

$$v_{orb} = \sqrt{\frac{\mu}{r}}$$

$$\Delta v_{deorbit} = v_{orb} \left(1 - \sqrt{\frac{2r_p}{r_p + r_a}} \right)$$

where v_{orb} is the orbital velocity of the circular orbit, r_p is the radius at perigee, and r_a is the radius at apogee.